CONTRACTING RATIONAL BRIDGES GIVES A CANONICAL MODEL WITH AMPLE CANONICAL SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the base-change paragraph (source lines 6526-6537) presents its argument (pullback of omega plus Gamma \otimes K = Gamma_K by flat base change, combined with the Proj canonicity argument above) and ends with 'Some details omitted.'; the ingredient f_*O_X = O_{X'} for a single contraction is delegated to the bridge-contraction statement (source lines 6514-6516); both pointers resolve inside the chapter and the two cited lemmas are registered inputs.

Editorial note. The label lemma-contracting-rational-bridges names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- No rational bridges means ample canonical sheaf. Same chapter, \label{lemma-no-rational-bridge-ample-genus-geq-2}. Proof lines 6508-6510: after contracting rational bridges until none are left, omega_{X'} is ample.
- Canonical pullback under a bridge contraction. Same chapter, \label{lemma-rational-bridge-canonical}. Proof lines 6512-6514: f^*omega_{X'} = omega_X follows from this lemma together with induction over the sequence of contractions.
- Structure sheaf pushes down under a single bridge contraction. Same chapter, Example \label{example-rational-bridge}. Proof lines 6514-6516: f_*O_X = O_{X'} because this is true for the contraction of a rational bridge (author's delegation).
- Proper ample schemes are Proj of the section ring. Morphisms, Lemma \label{morphisms-lemma-proper-ample-is-proj}. Proof lines 6520-6524: X' is the Proj of the direct sum of the spaces Gamma(X', omega_{X'}^{\otimes m}), which makes the morphism X -> X' completely canonical.
- Dualizing module and cohomology under base field extension. Same chapter, \label{lemma-sanity-check-duality}; Cohomology of Schemes, Lemma \label{coherent-lemma-flat-base-change-cohomology}. Proof lines 6526-6537: omega_{X_K} is the pullback of omega_X and Gamma(X, omega_X^{\otimes m}) \otimes_k K = Gamma(X_K, omega_{X_K}^{\otimes m}) by flat base change.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Prerequisite pointer invoked at proof line(s) 6536; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3M: sources/stacks-curves/pinned-original-source.tex, source lines 6201–6201. Prerequisite pointer invoked at proof line(s) 6500; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3P: sources/stacks-curves/pinned-original-source.tex, source lines 6433–6433. Prerequisite pointer invoked at proof line(s) 6509; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3N: sources/stacks-curves/pinned-original-source.tex, source lines 6367–6367. Prerequisite pointer invoked at proof line(s) 6513; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E32: sources/stacks-curves/pinned-original-source.tex, source lines 927–927. Prerequisite pointer invoked at proof line(s) 6531; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C6J: sources/stacks-chow/tags.txt, source lines 15299–15299. Prerequisite pointer invoked at proof line(s) 6525; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k of dimension 1 with $H^0(X, \mathcal{O}_X) = k$ having genus $g \geq 2$. Assume the singularities of X are at-worst-nodal and that X has no rational tails. Consider a sequence*

$$X = X_0 \to X_1 \to \ldots \to X_n = X'$$

of contractions of rational bridges (Example 0E3M) until none are left. Then $\omega_{X'}$ ample. The morphism $X \to X'$ is independent of choices and formation of this morphism commutes with base field extensions.

Proof. We proceed by contracting rational bridges until there are none left. Then $\omega_{X'}$ is ample by Lemma 0E3P.

Denote $f : X \to X'$ the composition. By Lemma 0E3N and induction we see that $f^*\omega_{X'} = \omega_X$. We have $f_*\mathcal{O}_X = \mathcal{O}_{X'}$ because this is true for contraction of a rational bridge. Thus the projection formula says that $f_*f^*\mathcal{L} = \mathcal{L}$ for all invertible $\mathcal{O}_{X'}$-modules $\mathcal{L}$. Hence

$$\Gamma(X', \omega_{X'}^{\otimes m}) = \Gamma(X, \omega_X^{\otimes m})$$

for all m. Since X' is the Proj of the direct sum of these by Morphisms, Lemma 0C6J we conclude that the morphism $X \to X'$ is completely canonical.

Let K/k be an extension of fields, then ω_{X_K} is the pullback of ω_X (Lemma 0E32) and we have $\Gamma(X, \omega_X^{\otimes m}) \otimes_k K$ is equal to $\Gamma(X_K, \omega_{X_K}^{\otimes m})$ by Cohomology of Schemes, Lemma 02KH. Thus formation of $f : X \to X'$ commutes with base change by K/k by the arguments given above. Some details omitted. □